

Roche limits

Y22 (2022) — English translation

Gravitational forces increase as the distance between bodies decreases. Therefore, one can observe how, with a strong approach, for example, liquid satellites and planets with a strong gravitational field, the shape of the satellites begins to change and they are ultimately destroyed. The Roche limit is the minimum possible radius of a circular orbit of a satellite R_{Roche} revolving around a planet, at which the satellite is not yet destroyed under the influence of tidal forces caused by the gravitational field of the planet. In this problem you have to find the Roche limits for a binary planetary system, as well as for a liquid satellite.

Part A. Roche limit for a binary planetary system (1.5 points)

Consider two identical spherical objects, the mass and radius of each being m and r , respectively. Objects as a whole move in the gravitational field of a star with mass $M \gg m$ in circular orbits. Let us denote the distance from the point of contact of the objects to the star as R . Consider that $R \gg r$ and the centers of the objects are always on a straight line passing through the centers of the star (that is, the objects always face one side to the star). The gravitational constant is G .

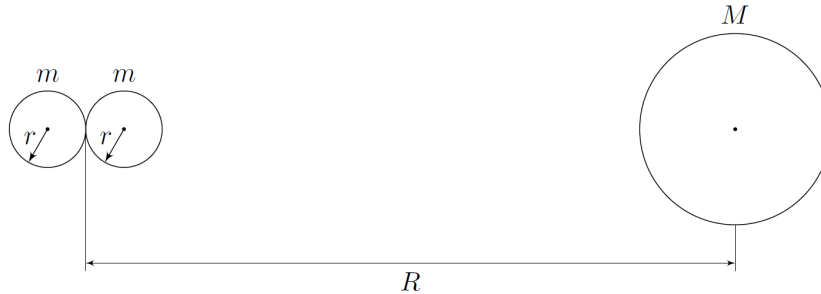


Figure 1: Figure 1.

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| A1 | Find the value of the radius of the circular orbit R_{BPS} at which the mechanical stress in the area of contact of the objects is equal to zero. The quantity R_{BPS} is the Roche limit R_{Roche} for a binary planetary system. | 1.50 |
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Liquid satellite

All remaining points of the problem lead to finding the Roche limit for a liquid satellite. rotating around its own axis synchronously with the orbital rotation (in other words, the satellite always faces the same side towards the planet.). Let us denote the radius of the satellite's circular orbit as R . The gravitational constant G , the radius of the planet R_0 , the density of the planet ρ_M and the density of the liquid satellite ρ_m are considered known at all points of the problem. The problem is solved in a reference frame rotating around the center of the planet with the orbital angular velocity of rotation of the satellite, and also under the following assumptions: 1) the center of the planet is stationary; 2) the influence of all forces, except centrifugal, gravity of the

planet and the satellite's own gravity can be neglected; 3) the liquid is incompressible; 4) in a rotating reference frame the fluid is motionless; 5) all linear dimensions of the satellite are much smaller than the radius of the circular orbit R . Also, when solving the problem, use a coordinate system, the origin of which is located at the position of the satellite's center of mass, the x axis is directed along a line drawn from the planet to the satellite, the z axis is directed along the angular velocity vector of orbital rotation, and the y axis complements the three basis vectors $\vec{e}_x, \vec{e}_y, \vec{e}_z$ to the right.

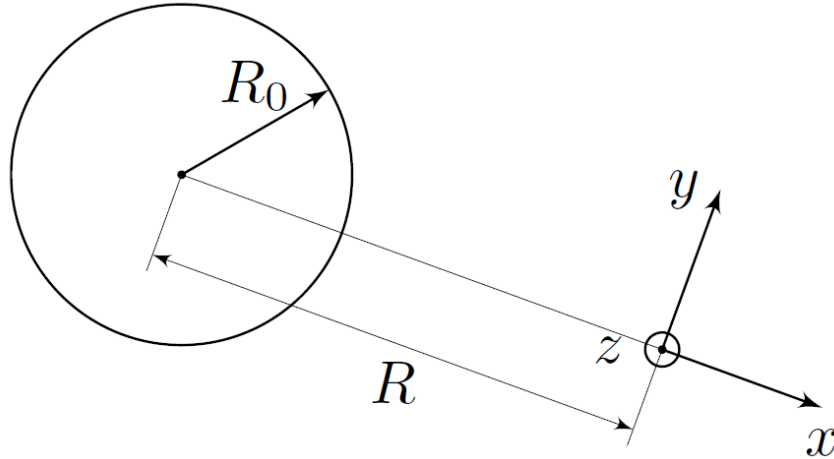


Figure 2: Figure 2.

Part B. Potential of external fields (1.5 points)

We will call the potential of a body the potential energy of a unit of mass, i.e.:

$$\varphi = \frac{W_p}{m}.$$

In this part of the problem, it is necessary to approximately describe the total potential φ_{ext} of the centrifugal force field and the gravitational field of the planet near the equilibrium position of the satellite. Assume that at the origin of coordinates $\varphi_{\text{ext}0} = 0$.

B1	Obtain an exact expression for the total potential φ_{ext} in the field of the centrifugal force and the gravitational field of the planet. Express your answer in terms of x, y, z, G, ρ_M, R_0 and R .	0.70
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B2	In the $x, y, z \ll R$ approximation, the expression for the potential φ can be represented as follows:	0.80
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$$\varphi_{\text{ext}} = A_1 x^2 + B_1 y^2 + C_1 z^2.$$

Find the coefficients A_1, B_1 and C_1 . Express your answers in terms of R, G, R_0 and ρ_M .

In what follows, assume that the coefficients B_1 and C_1 are equal to zero. As accurate calculations show, taking them into account gives a very small increase in accuracy, incomparable with an increase in the complexity of calculations. Thus, in all further points:

$$\varphi_{\text{ext}} = A_1 x^2$$

From the form of the potential it becomes clear that the figure we are considering is a figure of rotation, and the quadratic form of the dependence of the potential makes it possible to guess the geometric figure.

Part C. Gravitational field of an ellipsoid of revolution (5.5 points)

In this part of the problem, you are asked to familiarize yourself with the gravitational field inside an elongated ellipsoid of revolution, filled throughout its entire volume with a substance with density ρ . In the Cartesian coordinate system xyz the ellipsoid of revolution is described by the equation:

$$\frac{x^2}{a^2} + \frac{y^2 + z^2}{b^2} \leq 1, \quad a \geq b,$$

where a and b are, respectively, the semi-major and semi-minor axes of the ellipse in any cross-section containing the origin. Recall that the foci of the ellipse are located at a distance $2c = 2\sqrt{a^2 - b^2}$ from each other, and the eccentricity of the ellipse is $e = \frac{c}{a}$.

Let's consider an auxiliary problem: A cavity was created in the ellipsoid under consideration, specified by the equation:

$$\frac{x^2}{k^2 a^2} + \frac{y^2 + z^2}{k^2 b^2} \leq 1 \quad k < 1$$

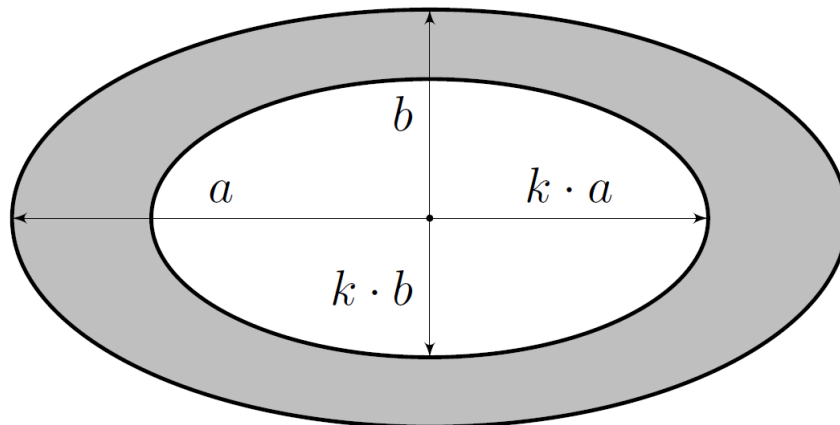


Figure 3: Figure 3.

C1 Prove that the gravitational field inside the cavity is zero.

0.80

Let's return to the solid ellipsoid.

C2 Let the potential at the center of the ellipsoid be φ_0 . Consider a point inside the ellipsoid, the radius vector relative to the center is equal to $\vec{r}$. The potential at this point is equal to φ_1 . What is the potential $\varphi_2 = \varphi(\gamma\vec{r})$ at the point with radius vector $\gamma\vec{r}$ also located inside the ellipsoid? Express your answer in terms of φ_0 , φ_1 and γ .

0.40

- C3** Show that at a point on an ellipsoid with coordinates x, y, z the potential can be represented as: **1.00**

$$\varphi = \varphi_0 + A_2 x^2 + B_2 (y^2 + z^2)$$

Note: note that this expression is applicable for ellipsoids of arbitrarily small sizes

- C4** Express B_2 in terms of A_2 , G and ρ . *Note:* use Gauss's theorem in differential form for the gravitational field: **0.30**

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 4\pi\rho G$$

- C5** The potential φ on the surface of the ellipsoid can be represented as a function of only the coordinate x in the following form: **0.30**

$$\varphi = C_2 + x^2 D_2.$$

Express C_2 and D_2 in terms of φ_0 , A_2 , G , ρ , a , x and the eccentricity e .

As can be seen, an ellipsoid of revolution can create a field that satisfies the condition of equipotentiality of the liquid surface. To do this, it is necessary to find the dependence of the coefficient A_2 on the eccentricity e of the ellipsoid.

- C6** Let us solve an auxiliary problem: mass with surface density σ is uniformly distributed over the surface of a disk of radius R . Find the potential of the disk φ_d on the axis at a distance x from its center. Express your answer in terms of σ , G , R and x . **0.50**

- C7** Find the potential φ_0 at the center of the ellipsoid. Express your answer in terms of G , ρ , a and e . *Note:* you may need the following integral: **1.00**

$$\int \sqrt{1+x^2} dx = \frac{x\sqrt{1+x^2} + \ln(x + \sqrt{1+x^2})}{2} + C$$

- C8** Find the potential of the ellipsoid φ_a at the point with coordinates $(x, y, z) = (a, 0, 0)$. Express your answer in terms of G , ρ , a and e . *Note:* you may need the following integral: **1.00**

$$\int \sqrt{x^2-1} dx = \frac{x\sqrt{x^2-1} - \ln(x + \sqrt{x^2-1})}{2} + C$$

- C9** Find the coefficient A_2 . Express your answer in terms of G , ρ and e . **0.20**

Part D. Finding the Roche limit (1.5 points)

For a liquid satellite to be in equilibrium, its surface must be equipotential. In the previous part of the problem, we showed that the potential on the surface of a homogeneous ellipsoid of revolution can be represented as a function of only the coordinate x .

D1	On a sheet of graph paper, plot the dependence of the coefficient D_2 on the eccentricity e of the ellipsoid formed by the satellite.	0.70
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D2	At what value of eccentricity e is the Roche limit reached?	0.30
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D3	Find the Roche limit R_{Roche} . Express your answer in terms of R_0 , ρ_M and ρ_m .	0.50
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Source: <https://pho.rs/p/2442>